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UNIT 1: Linux Basics

1. What is Linux?

Linux is an open-source operating system based on the Linux kernel. It manages hardware resources and provides services to applications and users.

Interview Answer:

Linux is an open-source operating system widely used in servers, cloud platforms, enterprise applications, and networking because of its stability, security, and performance.

2. What is an Operating System?

An Operating System (OS) is software that acts as an interface between the user and computer hardware.

Functions:

- Process Management
- Memory Management
- File Management
- Device Management

Examples: Linux, Windows, macOS

3. What is Linux Kernel?

The kernel is the core part of Linux.

Responsibilities:

- CPU management
- Memory management
- Process scheduling
- Hardware communication

Interview Answer:

The Linux kernel is the core component of Linux that manages hardware resources and acts as a bridge between software and hardware.

4. What is Open Source?

Open source means the source code is publicly available and can be modified and distributed by anyone.

Examples:

- Linux
 - MySQL
 - Apache
-

5. What are Linux Distributions?

A Linux distribution is a complete operating system built using the Linux kernel and additional software packages.

Examples:

- Ubuntu
- CentOS
- Red Hat Enterprise Linux (RHEL)

- Debian
-

6. What is Ubuntu?

Ubuntu is a popular Debian-based Linux distribution known for its ease of use and large community support.

7. What is CentOS?

CentOS is a Linux distribution derived from Red Hat Enterprise Linux and is commonly used on servers.

8. What is RHEL?

RHEL (Red Hat Enterprise Linux) is a commercial Linux distribution used by enterprises for stability, support, and security.

9. Difference Between Linux and Unix

Linux	Unix
Open Source	Mostly Proprietary
Free	Paid
Community Developed	Vendor Developed
Widely Used Today	Used in Some Enterprise Systems

Interview Answer:

Unix is a complete operating system family, while Linux is an open-source kernel used to build operating systems.

10. Why is Linux Popular?

- Open Source
 - Secure
 - Stable
 - Fast
 - Multi-user
 - Widely used in servers
-

11. What is CLI?

CLI (Command Line Interface) allows users to interact with Linux using commands.

Example:

```
ls  
pwd  
cd
```

12. What is GUI?

GUI (Graphical User Interface) allows users to interact through windows, icons, and menus.

Examples:

- Ubuntu Desktop
- GNOME

13. Difference Between CLI and GUI

CLI	GUI
Command Based	Graphical
Faster	Easier for beginners
Less Resource Usage	More Resource Usage

UNIT 2: Navigation Commands

1. What is Navigation?

Navigation means moving between directories and locating files in Linux.

2. pwd Command

Full Form

Print Working Directory

Purpose

Shows current directory.

Command:

```
pwd
```

Example Output:

```
/home/vivek
```

Interview Answer:

pwd displays the current working directory.

3. ls Command

Lists files and directories.

Command:

```
ls
```

4. ls -l

Shows detailed information.

Command:

```
ls -l
```

Displays:

- Permissions
 - Owner
 - File Size
 - Date
-

5. ls -a

Shows hidden files.

Command:

```
ls -a
```

Example:

```
.bashrc  
.profile
```

6. Difference Between ls, ls -l, ls -a

Command	Purpose
ls	List files
ls -l	Detailed list

ls -a	Show hidden files
-------	-------------------

7. cd Command

Full Form

Change Directory

Used to move between folders.

Example:

```
cd project
```

8. cd ..

Moves one level up.

Example:

Current:

```
/home/vivek/project
```

Command:

```
cd ..
```

Result:

```
/home/vivek
```

9. cd ~

Moves to Home Directory.

Command:

```
cd ~
```

or

```
cd
```

10. cd -

Moves to previous directory.

Command:

```
cd -
```

11. What is Home Directory?

Personal directory of a user.

Example:

12. What is Root Directory?

Top-most directory in Linux.

/

All files and directories originate from root.

13. What is Absolute Path?

Complete path starting from root.

Example:

/home/vivek/project

14. What is Relative Path?

Path based on current location.

Example:

Current:

/home/vivek

Command:

project

15. Difference Between Absolute and Relative Path

Absolute Path	Relative Path
Starts from /	Starts from current location
Full path	Partial path
Independent of current location	Depends on current location

UNIT 3: Files & Directories

1. What is a File?

A file is a collection of data stored in the system.

Examples:

- notes.txt
- report.pdf
- app.java

Interview Answer:

A file is a container used to store information such as text, images, programs, or configuration data.

2. What is a Directory?

A directory is a folder used to organize files and other directories.

Example:

```
project/  
├── index.html  
├── style.css  
└── images/
```

3. touch Command

Used to create an empty file.

```
touch test.txt
```

4. mkdir Command

Used to create a directory.

```
mkdir project
```

5. cat Command

Displays file contents.

```
cat test.txt
```

6. head Command

Displays first 10 lines of a file.

```
head log.txt
```

7. tail Command

Displays last 10 lines of a file.

```
tail log.txt
```

GreyOrange Important

```
tail -10 app.log
```

Used to check recent logs.

8. cp Command

Copies files.

```
cp source.txt destination.txt
```

9. Copy Directory

```
cp -r folder1 folder2
```

10. mv Command

Moves or renames files.

Move:

```
mv test.txt Documents/
```

Rename:

```
mv old.txt new.txt
```

11. rm Command

Deletes files.

```
rm test.txt
```

12. rm -r

Deletes directory and contents.

```
rm -r project
```

13. rm -rf

Force deletes directory.

```
rm -rf project
```

Very dangerous command.

14. Difference Between cp and mv

cp	mv
Copies file	Moves file
Original remains	Original removed

15. Difference Between rm and rm -r

rm	rm -r
Deletes file	Deletes directory recursively

16. What is a Log File?

A log file stores application and system events.

Examples:

```
app.log
error.log
access.log
```

17. Why are Log Files Important?

They help:

- Detect errors
- Troubleshoot issues
- Find root causes
- Monitor applications

Since GreyOrange is a Software Support role, logs are extremely important.

UNIT 4: Searching & Logs

1. What is grep?

grep is used to search text inside files.

Example:

```
grep "ERROR" app.log
```

Output:

Shows all lines containing ERROR.

Interview Answer:

grep is a command used to search for a specific pattern or text inside a file.

2. grep -r

Searches recursively inside folders.

```
grep -r "ERROR" logs/
```

3. What is find?

Used to search files and directories.

Example:

```
find . -name test.txt
```

Searches for test.txt in current directory and subdirectories.

4. Find Large Files

```
find . -size +100M
```

Finds files larger than 100 MB.

5. What is locate?

Quickly finds files using a database.

```
locate test.txt
```

6. Difference Between find and locate

find	locate
Searches in real time	Uses database
Slower	Faster
More accurate	May show old results

7. What is which?

Shows location of a command.

Example:

```
which python
```

Output:

```
/usr/bin/python
```

8. What is whereis?

Shows:

- Binary file
- Source file

- Manual page

Example:

```
whereis python
```

9. What is Log Analysis?

Log analysis means examining log files to identify errors and issues.

Example:

```
tail app.log
```

Check latest entries.

10. Searching Errors in Logs

Example:

```
grep ERROR app.log
```

Searches all ERROR entries.

11. Searching Warnings in Logs

```
grep WARNING app.log
```

12. How Support Engineers Use Logs

Suppose customer says:

Application is not working.

Steps:

1. Check latest logs

```
tail app.log
```

2. Search errors

```
grep ERROR app.log
```

3. Identify root cause

4. Resolve or escalate
-

13. What is Root Cause Analysis (RCA)?

RCA means identifying the actual reason behind an issue instead of fixing only the symptoms.

Example:

Application crashed.

Wrong approach:

- Restart application

Correct approach:

- Check logs
- Find memory issue
- Fix root cause

UNIT 5: Permissions

Permissions control who can read, write, or execute a file.

1. What are Linux Permissions?

Permissions define what actions a user can perform on a file or directory.

Types:

- Read (r)
- Write (w)
- Execute (x)

Interview Answer:

Linux permissions control access to files and directories by defining read, write, and execute privileges.

2. Read Permission (r)

Value = 4

Allows user to:

- Open file
- Read contents

Example:

r--

3. Write Permission (w)

Value = 2

Allows user to:

- Modify file
- Edit contents

Example:

-w-

4. Execute Permission (x)

Value = 1

Allows user to:

- Run file as a program

Example:

```
--x
```

5. User Categories

Owner (u)

Person who owns the file.

Group (g)

Users belonging to same group.

Others (o)

Everyone else.

6. Understanding Permission String

Example:

```
-rwxr-xr--
```

Breakdown:

Owner = rwx

Group = r-x

Others = r--

7. chmod Command

Used to change permissions.

Example:

```
chmod 755 file.sh
```

Interview Answer:

chmod is used to modify file and directory permissions.

8. chmod +x

Makes file executable.

```
chmod +x script.sh
```

This command is specifically mentioned in GreyOrange's preparation material.

9. What is 777?

`rwX rwX rwX`

Everyone gets:

- Read
- Write
- Execute

Command:

`chmod 777 file.txt`

Not recommended for security.

10. What is 755?

`rwX r-X r-X`

Owner:

- Read
- Write
- Execute

Others:

- Read
- Execute

Command:

`chmod 755 file.sh`

Most common permission.

11. Difference Between 755 and 777

755	777
More Secure	Less Secure
Only owner can write	Everyone can write
Commonly used	Rarely used

12. chown Command

Changes file owner.

Example:

`chown vivek file.txt`

13. Change Owner and Group

```
chown vivek:developers file.txt
```

14. Why Use chown?

Used when:

- Ownership needs changing
 - User migration
 - Permission issues
-

15. Common Interview Scenario

Question:

File is not executing. What will you do?

Answer:

```
chmod +x file.sh
```

16. Another Scenario

Question:

Wrong owner is assigned to a file.

Answer:

```
chown username filename
```

UNIT 6: Process & Resource Monitoring

Very important for Software Support Engineers.

1. What is a Process?

A process is a program currently running in memory.

Examples:

- Chrome
- MySQL
- Apache
- Java application

Interview Answer:

A process is an instance of a running program.

2. What is PID?

PID = Process ID

Each process gets a unique number.

Example:

PID = 1024

3. ps Command

Shows running processes.

ps

Interview Answer:

ps displays information about active processes.

4. top Command

Shows:

- CPU usage
- Memory usage
- Running processes

top

5. htop Command

Improved version of top.

htop

Easier to read.

GreyOrange email specifically mentions htop.

6. Difference Between top and htop

top	htop
Basic	Advanced
Less Interactive	Interactive
Text Based	Better UI

7. kill Command

Stops a process.

```
kill PID
```

Example:

```
kill 1234
```

8. killall Command

Stops all processes with same name.

```
killall chrome
```

9. pidof Command

Finds process ID.

```
pidof mysql
```

Output:

```
1234
```

10. free -m Command

Shows RAM usage in MB.

```
free -m
```

This command is specifically mentioned by GreyOrange.

11. free -h Command

Shows RAM in human-readable format.

```
free -h
```

Output:

```
2GB
```

```
4GB
```

```
8GB
```

12. uname -a

Shows system information.

```
uname -a
```

Displays:

- Kernel version
 - Architecture
 - Hostname
-

13. hostname Command

Displays machine name.

```
hostname
```

14. uptime Command

Shows:

- System running time
- Load average

```
uptime
```

15. Resource Monitoring

Support Engineers monitor:

CPU

```
top
```

```
htop
```

Memory

```
free -m
```

Processes

```
ps
```

16. Common Support Scenario

Question:

Application is slow.

What will you check?

Answer:

```
top
```

```
htop
```

```
free -m
```

Check:

- CPU usage
 - Memory usage
 - Running processes
-

17. Another Scenario

Question:

Application has hung.

Answer:

1. Find PID

`pidof app`

2. Kill process

`kill PID`

3. Restart application

UNIT 7: File Transfer & Remote Access

This unit is very important because GreyOrange's email specifically mentions **scp** and support engineers often work on remote servers.

1. What is Remote Access?

Remote access means connecting to another computer/server over a network and managing it without physically being present.

Interview Answer:

Remote access allows a user to connect to and manage a remote system over a network.

2. What is SSH?

Full Form

Secure Shell

SSH is a secure protocol used to connect to remote Linux servers.

Command:

`ssh user@server_ip`

Example:

`ssh vivek@192.168.1.10`

Interview Answer:

SSH is a secure protocol used for remote login and administration of Linux servers.

3. Why is SSH Used?

- Secure communication
- Remote server management

- Encrypted connection
 - System administration
-

4. What is SCP?

Full Form

Secure Copy Protocol

Used to securely transfer files between systems.

Command:

```
scp file.txt user@server:/path
```

Example:

```
scp app.log vivek@192.168.1.10:/home/vivek
```

This command is directly mentioned in GreyOrange's preparation email.

5. Why Use SCP?

- Secure file transfer
 - Uses SSH encryption
 - Common in Linux servers
-

6. What is SFTP?

Full Form

Secure File Transfer Protocol

Used for interactive file transfer between local and remote systems.

Command:

```
sftp user@server
```

7. Difference Between SCP and SFTP

SCP	SFTP
Single file transfer	Interactive session
Faster	More flexible
One command	Multiple file operations

8. What is SSH Key Authentication?

Instead of passwords, SSH can use key pairs.

Benefits:

- More secure
- Easier automation
- Preferred in enterprises

Command:

```
ssh-keygen
```

9. Real Support Engineer Scenario

Question:

A customer sends a log file. How would you copy it to a server?

Answer:

```
scp app.log user@server:/path
```

UNIT 8: Networking Basics

Support Engineers frequently troubleshoot network issues.

1. What is a Network?

A network is a group of connected devices that communicate with each other.

Examples:

- Internet
 - Office LAN
 - Data Center Network
-

2. What is an IP Address?

An IP address uniquely identifies a device on a network.

Example:

```
192.168.1.10
```

Interview Answer:

An IP address is a unique numerical address assigned to a device on a network.

3. What is Hostname?

Hostname is the name assigned to a computer/server.

Example:

web-server

database-server

Command:

hostname

4. What is DNS?

Full Form

Domain Name System

DNS converts domain names into IP addresses.

Example:

google.com → 142.250.xxx.xxx

5. What is ping?

Used to test connectivity between systems.

Command:

ping google.com

Interview Answer:

ping checks whether a remote host is reachable over a network.

6. Why Use ping?

- Test connectivity
 - Check network availability
 - Measure response time
-

7. What is ifconfig?

Displays network interface information.

Command:

ifconfig

Shows:

- IP Address
 - Network Interface
 - MAC Address
-

8. What is ip addr?

Modern replacement for ifconfig.

Command:

```
ip addr
```

or

```
ip addr show
```

9. Difference Between ifconfig and ip addr

ifconfig	ip addr
Older command	Modern command
Less features	More features
Deprecated in many systems	Recommended

10. What is nslookup?

Used to find DNS information.

Command:

```
nslookup google.com
```

11. Why Use nslookup?

- Verify DNS resolution
 - Troubleshoot domain issues
 - Find IP addresses
-

12. What is localhost?

Local machine address.

```
127.0.0.1
```

Represents the current system.

13. What is a Port?

A port is a communication endpoint used by applications.

Examples:

Service	Port
HTTP	80
HTTPS	443
SSH	22

14. Real Support Engineer Scenario

Question:

Customer says application is not accessible.

Steps:

1. Check connectivity

```
ping server_ip
```

2. Check server IP

```
ip addr
```

3. Verify DNS

```
nslookup domain.com
```

UNIT 9: Shell Basics

Shell is the command interpreter that accepts commands from the user and sends them to the operating system for execution.

1. What is Shell?

A shell is a program that acts as an interface between the user and the Linux operating system.

Interview Answer:

A shell is a command-line interpreter that allows users to interact with the Linux operating system.

Examples:

- Bash
 - Sh
 - Zsh
 - Ksh
-

2. What is Bash?

Full Form

Bourne Again Shell

Most commonly used Linux shell.

Interview Answer:

Bash is the default shell in most Linux distributions and is used to execute commands and scripts.

3. What is a Shell Command?

A command entered into the terminal to perform a task.

Examples:

```
ls  
pwd  
mkdir
```

4. What is Alias?

An alias is a shortcut for a command.

Example:

```
alias ll='ls -l'
```

Now:

```
ll
```

acts as:

```
ls -l
```

5. Why Use Alias?

- Save time
- Simplify long commands
- Improve productivity

6. How to View Existing Aliases?

```
alias
```

7. How to Remove an Alias?

```
unalias ll
```

8. What is Command History?

Linux stores previously executed commands.

Command:

```
history
```

9. Why is History Useful?

- Reuse commands
 - Troubleshooting
 - Productivity
-

10. What is Man Command?

Full Form

Manual

Displays command documentation.

Example:

```
man ls
```

11. Why Use Man?

To learn:

- Command syntax
 - Options
 - Examples
-

12. What is Environment Variable?

A variable that stores system information.

Examples:

```
PATH
```

```
HOME
```

```
USER
```

13. What is PATH Variable?

PATH tells Linux where to find executable programs.

Check:

```
echo $PATH
```

14. What is HOME Variable?

Stores user's home directory.

Check:

```
echo $HOME
```

Example Output:

15. What is Echo?

Used to display text or variable values.

Example:

```
echo Hello
```

Example:

```
echo $HOME
```

16. Real Interview Question

Question:

What is alias and why is it used?

Answer:

Alias is a shortcut for a command. It helps reduce typing and improves productivity.

UNIT 10: Support Engineer Troubleshooting

This is the most important unit for GreyOrange because the role is Software Support.

1. What is Troubleshooting?

Troubleshooting is the process of identifying, analyzing, and resolving technical issues.

Interview Answer:

Troubleshooting is the systematic process of finding the root cause of a problem and resolving it.

2. What is Technical Support?

Technical support involves helping users and customers resolve software, hardware, and system issues.

3. What is Root Cause Analysis (RCA)?

RCA means identifying the actual reason behind a problem instead of only fixing symptoms.

Example:

Problem:

Application crashed.

Bad solution:

Restart application.

Good solution:

Find why it crashed.

4. What is an Incident?

An incident is an unexpected event that affects normal system operation.

Examples:

- *Server down*
 - *Application crash*
 - *Database connection failure*
-

5. What is Escalation?

Escalation means forwarding an issue to a higher-level team when it cannot be resolved.

Example:

Support Engineer → Development Team

6. What is a Ticket?

A ticket is a record of a customer issue.

Contains:

- *Problem details*
 - *Priority*
 - *Status*
 - *Resolution*
-

7. What is SLA?

Full Form

Service Level Agreement

Defines the expected response and resolution time.

Example:

Response Time: 30 minutes

8. Customer Reports Application Down

What will you do?

Step 1

Gather information.

Ask:

- What error are you seeing?
- Since when?
- Which application?

Step 2

Check logs.

```
tail app.log
```

Step 3

Search errors.

```
grep ERROR app.log
```

Step 4

Check server resources.

```
top
```

```
free -m
```

Step 5

Resolve or escalate.

9. Application is Slow

What will you check?

CPU Usage

```
top
```

```
htop
```

Memory Usage

```
free -m
```

Logs

```
tail app.log
```

10. File Permission Issue

Scenario:

Application cannot access a file.

Check:

```
ls -l
```

Fix:

```
chmod
```

```
chown
```

11. Log Investigation

Most common Software Support activity.

Commands:

```
tail
```

```
grep
```

```
cat
```

```
less
```

Goal:

- *Find errors*
 - *Understand issue*
 - *Identify root cause*
-

12. Customer Communication

When customer is frustrated:

1. *Stay calm*
2. *Listen carefully*
3. *Gather details*
4. *Give regular updates*
5. *Avoid blaming anyone*

Interview Answer:

I would remain professional, understand the issue, communicate clearly, provide updates, and work towards resolution.

13. Why Communication is Important?

GreyOrange specifically mentions client-facing support.

Good communication:

- *Builds trust*

- *Improves customer satisfaction*
 - *Reduces confusion*
-

14. Why Linux and SQL are Important?

Linux:

- *Log analysis*
- *Server monitoring*
- *Application support*

SQL:

- *Data investigation*
- *Database troubleshooting*
- *Query execution*